

Nathan Phillips

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Data Scientist | Applied AI Engineering

PROFESSIONAL SUMMARY

Data professional with 10+ years of experience delivering end to end data projects across startups/ tech / consulting and corporate. I generate business value by reverse-engineering outcomes leveraging skills in data engineering, analytics, machine learning and applied AI.

Currently upskilling in AI engineering pivoting to agentic coding to build and deliver my work and expand my impact from data projects to automations/ user facing projects.

KEY SKILLS

- **Data Science:** product analytics, opportunity sizing, experimentation, executive reporting, insight storytelling
- **Data Engineering:** ETL / ELT pipelines, AirFlow orchestration, Data Governance / Quality
- **Machine learning:** predictive modelling, segmentation, NLP, time series analysis, anomaly detection
- **AI engineering:** agentic coding, AI agents & RAG systems, evaluation-driven optimisation
- **Programming & Platforms:** SQL, Python, Javascript, R, PySpark. AWS / GCP / Databricks. Docker / Git

EXPERIENCE

AI Engineering, Career Development (Personal) | March 2026 - Current

Build this out more pull up the tools used

Full-time upskilling on AI and build agents to demonstrate experience and boilerplate code to leverage in the future.

- **Agentic Coding:** leverage coding-agent to build AI / Software systems through spec driven development. See `ai\_spec` folder for all below projects to see multiple spec for agent to implement.
- **Multi-agent system:** AI assistant answering multi-turn Question and Answers based financial reports doing both retrieval and calculation. <https://github.com/nmp-dsci/ConvFinQA-agent>
- **RAG Agent:** Legal agent extracting clauses from contract to review vs manually reviewing contract. Benchmarked different RAG techniques against evaluation set to optimise performance. <https://github.com/nmp-dsci/CUAD-agent>
- **Transcript Agent:** Q&A agent on YouTube transcripts using recursive RAG search to deep dive on topics. <https://github.com/nmp-dsci/yt-agent>
- **Data Agent:** Q&A agent for self-serve reporting, analytics and insights. TBC

Data Scientist, Atlassian | Sep 2022 - Mar 2026

Product Data Scientist; Bitbucket (May 2024 - Apr 2026) & Jira Work Management (Sep 2022 - May 2024).

Summarise tech and broad impact

- 3+ years experience in a fast paced environment with concurrent priorities across multiple teams for 1M MAU / \$100M ARR products. I lead tracking an improvement of SaaS products through routine business performance reporting, setting and measuring targets to hit FY goals, deep dives into opportunities to unlock growth, price rises, new edition offerings, experimentation to measure feature performance.
- **Unlock dedicated sales team** for Bitbucket Cloud migrations. By modelling historical Data Center customers for forward looking Cloud migrations, identified Jira Cloud migrators with Salesforce Bitbucket migration engagement are 3x more likely to migrate. Recommended ramp up of dedicated Sales team with customer list to target for migrations.
- **Funnel optimisation:** Identified that non-technical teams in Jira have higher paid conversion when they use a business templates, leading to onboarding flow experiment to show templates based on admin's craft (marketing / finance / business) leading to significantly uplift in signup activation for metrics D1T6 / W2WAI and M2 activity.
- **Data pipeline:** Jira Work Management (JWM) revenue comes through Jira paid seats (implicit: free via another paid product) or JWM paid seats (explicit). I worked with leadership to define and deliver a data pipeline to capture JWM's total revenue by using Databricks to build an ETL and dashboards to track actual vs set targets for the business. This project became the source of truth for product performance, attribution, insights and reporting

Lead Data Scientist, StockPay | Apr 2022 - Oct 2022

Founding data hire in pre-launch stock trading up offering buy now pay later. Went into administration pre-launch.

- **Architected analytics platform;** using AWS & terraform I built a serverless architecture; Lambda for API ingestion and ETL scheduling > S3 for raw storage > Athena for meta data and data warehouse > Quicksight for visualisation.
- Lead data instrumentation with engineering; implemented of google analytics, braze marketing, KYC, buy/sell transactions, portfolio tracking, buy now pay later.

Senior Data Scientist, NRMA | Apr 2019 - Apr 2022

Leading Data Science function with 1 direct report for a portfolio of businesses; 2M members, 200k breakdown callout business, Thrift car rentals, Holiday parks and IAG Insurance cross sell.

- **Source of truth data pipeline:** I built the source of truth for membership renewals business in PySpark (& later in GCP) consolidating multiple teams logic used for reporting / analytics and predictive modelling into 1 table. I lead the build, alignment with stakeholders and adoption so 3 data teams had the same numbers and shifted the effort from validation to value generation.
- **Predictive modelling:** I owned the predictive models for membership churn & cross sell of insurance, rentals & holiday parks ; with a 2m membership base churn was the no.1 initiative, outreach based on churn segments used in automated / call center outreach lead to lower churn vs holdout segments that didn't receive outreach.
- **Customer segmentation:** during Covid Thrifty car rental business experienced 50%+ declines in bookings sparking a war room for vehicle planning by branch. I scoped and delivered the segmentation bookings by branch, customer & vehicle booking to capture which branches had the biggest exposure to declines and the profile of branches by use case commercial / moving house / interstate / international / local. This segmentation guided the vehicle planning by branch at a time Thrifty rebranded to Sixt and understood the long term exposure of Covid on the business.

Lead Data Scientist, Ubicar Insurance | Feb 2019 - Apr 2019

Founding Data Scientist; built data platform in AWS; Lambda > S3 > Redshift > Quicksight for company pricing car Insurance based on telemetry data capture on app. The company went into Administration.

Data Scientist, Hyperanna | Feb 2018 - Feb 2019

Data scientist embedded in an engineering team producing automated insights on customer loaded data.

- **Anomaly detection:** Built automated insight generation for a web application using NLP to identify and rank the most interesting insights to server users with drill down to contributions to result using Twitter STL algorithm.
- **Insight generation optimisation:** rewrote anomaly detection algorithm in SQL to run where data is stored lead to 10x improvement and transition backend to Airflow and Docker to improve observability, reliability and performance tracing.

Customer Insights Analyst, Allianz | Apr 2017 - Apr 2018

Insights analyst dedicated to Allianz's insurance partners, I worked with partner banks / dealerships and mutuals to increase insurance policies sold through insights and data driven campaign list generation.

- **Cross sell model fusing Banking and Insurance data:** matched bank insurance transactions with insurance policy data to predict competitor transaction product and renewal date of the bank transaction for targeted cross sell campaigns. Campaign was significant for both natural conversion rate and marketing uplift especially when Allianz was expected to have better price than competitor leading to recurring campaigns.

Data Analyst, Quantum | Dec 2014 - Apr 2017

Member of the external consulting team dedicated to Qantas & owned 3x online purchases reports. I was recognised as a modelling specialist who supported predictive modelling for CoreLogic / Masters Home Improvement and Woolworth.

+20% improvement in segmentation classification accuracy:

Qantas conducted a 10k survey to classify loyalty members into a 8 class segmentation to capture premium-ness (JetStar vs Qantas economy - first class) and score this segmentation out to their 10m loyalty member base. Qantas internally could only achieve 50% accuracy, I was able to achieve 80% accuracy with exact same data.

How? By stacking multiple techniques to overcome the bias in survey and imbalances between segments:

- **Remove bias:** survey was conducted in Airport so bias to flyers; no inactive flyers. Implemented PCA on the 8m base as predictors to train a model on a 10k survey set. PCA's indexed the survey customers vs loyalty base enabling the model to recognise the survey set as active flyers.
- **Predicting the segmentation scale:** segments classify members as Jetstar to Qantas First Class flyers. By adding a model scoring the numeric value of the segment it anchored the segmentation to stop premium customers scoring on lower tier segments.
- **Stacking models:** Stack different models prediction: Random Forest / GBM / Linear score lead to final accuracy score.

Statistical Analyst, Nielsen Audience Measurement | Aug 2013 - Dec 2014

Help maintained a 5,000 household panel scaled to report on Australian TV viewing behaviour, supporting official TV ratings for networks, programs and advertisers. Delivered stratified sampling and predictive modelling to understand viewing behaviour and skews in panel to inform life cycle management.

EDUCATION

